UNCONDITIONAL EXCEPTIONAL-PARAMETER DENSITY BOUNDS FOR RAMIFIED GL_4 AUTOMORPHIC REPRESENTATIONS

ARGUS OPEN RESEARCH PROJECT

Abstract. Toward the Generalized Ramanujan Conjecture for cuspidal automorphic representations on GL_4, we establish an unconditional density theorem for representations in a ramified family $\mathcal{F}(q)$ whose Satake parameters violate temperedness by $t_\pi \geq \tau$. By constructing an augmented positive pretrace test weight that repairs the vanishing of individual identity Whittaker functionals on conjugate ramified types and bounding all seven nonidentity orbital exclusions, we prove that $\#\{\pi \in \mathcal{F}(q) : t_\pi \geq \tau\} \ll_\varepsilon q^{3-4\tau+\varepsilon}$ uniformly for $0 < \tau \leq 1/8$.

1. Main Mathematical Results

Theorem 1.1 (Unconditional ramified-type density estimate on GL_4)**.** *For $0 < \tau \leq 1/8$ and every $\varepsilon > 0$, the number of cuspidal automorphic representations $\pi \in \mathcal{F}(q)$ on GL_4 with non-temperedness parameter $t_\pi \geq \tau$ satisfies*

$$\#\big\{\pi \in \mathcal{F}(q) : t_\pi \geq \tau\big\} \ \ll_\varepsilon \ q^{3-4\tau+\varepsilon}.$$

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 9400078 in the domain of automorphic forms and analytic number theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Unconditional Sarnak-type density bound « q^(3-4*tau+epsilon) (0 < tau <= 1/8) for non-tempered ramified automorphic representations in the GL_4 family.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted Generalized Ramanujan Conjecture t_pi = 0 for all cuspidal automorphic representations of GL_n(A_K)

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 9400078 (automorphic forms and analytic number theory), archived in the accompanying repository package `results/9400078/`.

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